Collocations in Emma (PMI)

Step 1: Install NLTK

Open Command Prompt or Terminal:

```
pip install nltk
```

Step 2: Download Required Corpora

Run Python and execute:

```
import nltk

nltk.download('gutenberg')
nltk.download('wordnet')
nltk.download('omw-1.4')
```

Step 3: Create the Program

Save the following code as collocations_wordnet.py:

```
import nltk
from nltk.collocations import (
    BigramCollocationFinder,
    BigramAssocMeasures
)
from nltk.corpus import gutenberg, wordnet as wn

# ------------------------------------------------
# Task 1: Collocations in Emma (PMI)
 # ------------------------------------------------

 words = [
     w.lower()
     for w in gutenberg.words("austen-emma.txt")
     if w.isalpha()
 ]

 finder = BigramCollocationFinder.from_words(words)

 # Ignore rare bigrams
```

```
finder.apply_freq_filter(5)

print("Task 1: Collocations in Emma (PMI)")
print("-" * 40)

for i, bg in enumerate(
        finder.nbest(BigramAssocMeasures.pmi, 10), start=1):
    print(f"{i}. {' '.join(bg)}")

# ---------------------------------------------------
# Task 2: WordNet senses of car
# ---------------------------------------------------

print("\nTask 2: WordNet Senses of 'car'")
print("-" * 40)

for syn in wn.synsets("car")[:3]:

    print("\n", syn.name())

    print("Synonyms:")
    print(", ".join(syn.lemma_names()))

    print("Hypernyms:")
    for h in syn.hypernyms():
        print("   ", h.name())
```

Step 4: Run the Program

```
python collocations_wordnet.py
```

Expected Output

PMI Collocations

```
Task 1: Collocations in Emma (PMI)

1. barouche landau
2. o clock
3. court plaister
```

```
4. brunswick square
5. sore throat
6. d ye
7. don t
 8. mill farm
 9. william larkins
 10. baked apples
```

These word pairs are **collocations**, meaning they frequently occur together in *Emma*.

Examples:

```
sore throat
baked apples
```

These are examples of **syntagmatic relations** because the words co-occur.

WordNet Senses of "car"

```
car.n.01
Synonyms:
car, auto, automobile, machine, motorcar

Hypernyms:
motor_vehicle.n.01
```

```
car.n.02
Synonyms:
car, railcar, railway_car, railroad_car

Hypernyms:
wheeled_vehicle.n.01
```

```
car.n.03
Synonyms:
car, gondola

Hypernyms:
compartment.n.02
```

These are examples of **paradigmatic relations**, where words belong to the same semantic class and can often substitute for one another.

Lab Record Result

Syntagmatic Relations

The PMI analysis on the **Emma** corpus produced collocations such as:

```
sore throat
baked apples
barouche landau
```

These words show strong co-occurrence patterns.

Paradigmatic Relations

The **WordNet** analysis shows multiple meanings of the word **car**:

1. Automobile
2. Railway car
3. Gondola/compartment

along with their hypernyms:

```
motor_vehicle
wheeled_vehicle
compartment
```

Conclusion

This experiment successfully demonstrates:

1. **Syntagmatic Relations** using PMI-based collocation extraction from the *Emma* corpus.
2. **Paradigmatic Relations** using WordNet synsets and hypernyms for the word **car**.

These concepts are fundamental in **Natural Language Processing (NLP)** for semantic analysis, collocation extraction, text mining, and language understanding.